Weekly Business Review — Olist E-commerce Marketplace

Objective: Build a reproducible data-analysis workflow using Olist marketplace CSV data and PostgreSQL.

1. Data Pipeline

Nine raw CSV datasets are loaded into PostgreSQL. Staging tables standardize orders, customers, items, payments and reviews. The analytics.order_mart provides an order-level view for business analysis. Python handles ingestion, quality checks and chart generation.

2. Data Quality Results

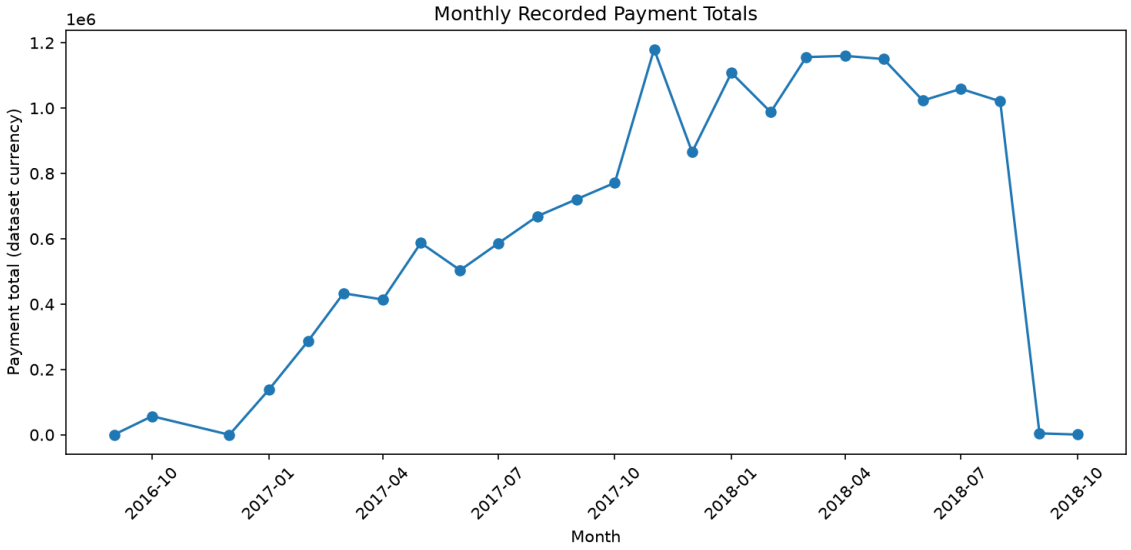
Check	Observed failures
Duplicate order IDs	0
Null order IDs	0
Delivery before purchase	0
Missing customer records	0
Payment mismatches	1,022
Invalid review scores	0

Payment mismatches were flagged for investigation using a tolerance of 1.00 between payment totals and item-plus-freight totals. Flagged records should not be automatically deleted.

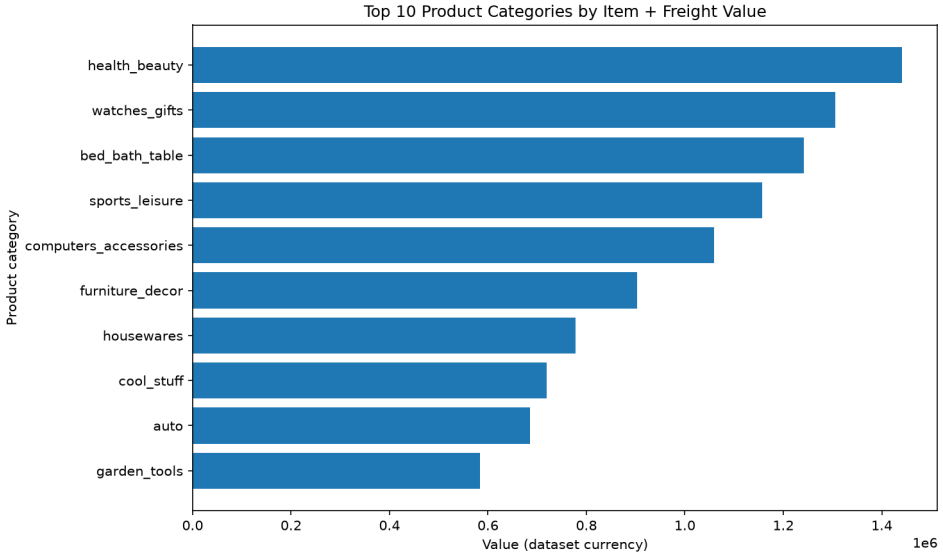
The mart contains 99,441 orders and 98,673 orders with review scores.

3. Revenue and Product Category Analysis

Monthly recorded payment totals rise across much of the observed period, with fluctuations in later months. The final months may be incomplete; therefore, the sharp final decline must not be interpreted as a confirmed business downturn.



The category chart ranks product categories by item price plus freight value. Health and beauty is the highest-valued category in the generated chart. This measure is not profit.

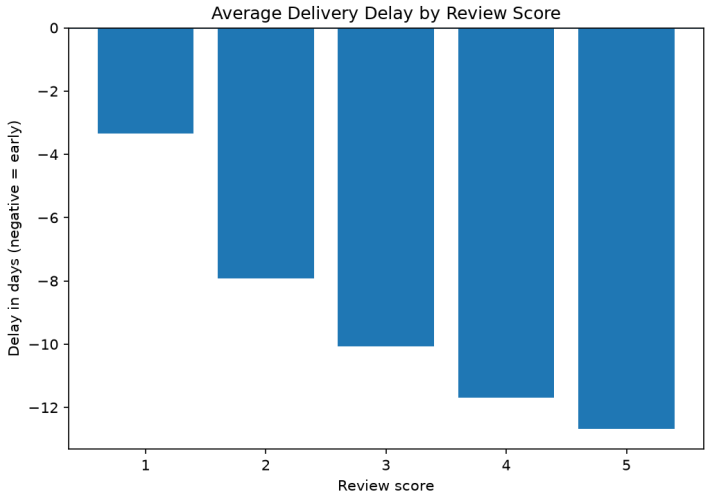


4. Customer Reviews and Delivery Performance

Review scores provide a view of customer feedback. The distribution chart should be used to identify which scores are most frequent.



Average delivery delay is calculated as actual delivery date minus estimated delivery date. Negative values indicate delivery before the estimated date; positive values indicate lateness.



The observed relationship between delivery timing and review score is associative and does not establish causation.

5. SQL Analysis

Monthly revenue: Uses LAG to compare monthly payment totals and calculate month-over-month growth.

Category performance by state: Uses DENSE_RANK to rank product categories within each customer state.

Order trend: Calculates a seven-day rolling average of daily order counts. Calendar-day gaps should be considered when interpreting a row-based rolling window.

Repeat purchases: Uses customer_unique_id to identify customers with more than one order. The observed repeat-purchase rate is 2,997 / 96,096 = 3.12%.

Delivery and reviews: Compares average delivery delay with review scores to explore a possible relationship.

6. VP Memo

To: Marketplace Leadership

Subject: Customer retention, delivery experience and payment quality

The analysis highlights three areas for action. First, the repeat-purchase rate is 3.12%, so customer retention merits further investigation. Second, review scores and delivery timing should be monitored together to identify service improvements. Third, 1,022 payment mismatches require reconciliation and investigation.

Recommendations

1. Investigate payment mismatches by order status and transaction details.

2. Monitor delivery delay and review-score trends together.

3. Explore customer retention by cohort and purchase frequency.

4. Annotate incomplete months in revenue reports and avoid unsupported conclusions about recent declines.

Limitations

The findings depend on the available dataset and implemented checks. Payment mismatches are flagged, not proven errors. Observational associations do not prove causation. The pipeline is a local project implementation, not a scheduled production service.